When Common Optimal Projections Exist: A Trace/Operator-Norm Fork for PSD Families

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Abstract

For a finite family of positive semidefinite operators on $\mathbb{R}^d$, we compare two notions of common rank-r optimality: the Ky Fan trace score $\mathrm{tr}(PM)$ and the operator-norm score $\lambda_{\max}(PKP)$. For the trace score, a common optimal projection exists exactly when the family has a shared dominant r-dimensional invariant subspace (*r-coherence*). For the operator-norm score, a common optimal projection exists exactly when one r-plane intersects every leading eigenspace (*r-coverability*). These conditions differ: r-coherence implies r-coverability, but the converse fails, as shown by a 3×3 witness. When coverability fails, we introduce an eigenvalue-weighted Ky Fan surrogate and prove coarse and instance-dependent regret bounds. The results clarify how the existence of a common optimal low-rank projection depends on the scoring rule.

Keywords: PSD families, finite-rank projection, Grassmannian, Ky Fan theorem, common principal subspace, operator-norm, r-coherence, r-coverability, regret bound
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1. Introduction

A recurrent structural question in low-rank approximation, sensor / output selection, and reduced-order modelling is whether a single rank-r orthogonal projection can be simultaneously optimal for an entire finite family of positive semidefinite (PSD) targets, or whether the optimal projection is inherently task-dependent. The answer depends not only on the family but on *which scoring rule* measures projection quality.

For fixed rank r on $\mathbb{R}^d$, two PSD-objective scoring rules are natural:

- the *trace* score $S_t^{\mathrm{tr}}(P) = \mathrm{tr}(PM_t)$, governed by the Ky Fan variational principle [1];
- the *operator-norm* score $S_t^{\mathrm{op}}(P) = \lambda_{\max}(PK_tP)$, governed by the Rayleigh quotient characterisation of the largest eigenvalue.

The trace score corresponds to squared-error prediction risk in linear Gaussian settings (§3); the operator-norm score admits a linearised Jacobian-response motivation (§4). The individual variational steps are classical [1, 2]. The contribution of this note is the comparison: the trace objective requires a common dominant r-subspace, whereas the operator-norm objective only requires coverability of leading eigenspaces. The strict separation between these existence laws (Theorem 21), and the surrogate regret analysis beyond coverability (§6), are the main new components.

Main results.

1. *Trace-score characterisation* (Proposition 4): an exact characterisation by Ky Fan trace maximisation. A common optimal rank-r projection for $\{M_t\}$ under $\mathrm{tr}(PM_t)$ exists iff the family is *r-coherent*, i.e. all M_t share a common dominant r-dimensional invariant subspace. The squared-error linear Gaussian dynamical systems (LGDS) Bayes-risk instance is recovered as a corollary.

Table 1: Side-by-side comparison of the two scoring rules considered in this paper. The trace score is smooth (analytic) on $\mathrm{Gr}(r, d)$; the operator-norm score is continuous, smooth only on the stratum where the leading eigenvalue of PKP is simple (§2).

Scoring rule	Fixed-target optimal P has	Common optimum exists iff
$\mathrm{tr}(PM)$ (Ky Fan trace)	range = a dominant r-dim invariant subspace of M	family is r-coherent (Prop. 4)
$\lambda_{\max}(PKP)$ (operator norm)	range $\cap\, E_{\max}(K) \neq \{0\}$	family is r-coverable (Prop. 17)

2. *Operator-norm characterisation* (Proposition 17): an exact characterisation by the rank-1-driven structure of the operator-norm maximiser. A common optimal rank-r projection for $\{K_t\}$ under $\lambda_{\max}(PK_tP)$ exists iff the family is *r-coverable*, i.e. some rank-r subspace intersects $E_{\max}(K_t)$ nontrivially for every t. The operator-norm score has a linearised Jacobian-response motivation (§4); this is a local linear-algebra setting, not statistical causal identification.
3. *The fork* (Theorem 21): for a single PSD family, r-coherence implies r-coverability strictly. A 3×3 witness shows the gap: a single common optimal projection can exist under the operator-norm score while no common optimal projection exists under the trace score; in the witness, the trace-score minimax regret is exactly $\Gamma_{\mathrm{tr}}^{\mathrm{fork}} = 2$ (§7), so the fork is not merely existential but has a finite minimax gap under the trace objective. The dissociation is algebraic, not a consequence of approximation error.
4. *Surrogate regret beyond coverability* (Theorems 23, 24): when coverability fails, the eigenvalue-weighted surrogate $B_\lambda = \sum_i w_i \lambda_1(A_i) u_i u_i^\top$ converts the non-Ky–Fan operator-norm objective into a Ky–Fan eigenproblem. The surrogate is sandwiched between F and $F + \sum_i w_i \lambda_2(A_i)$, giving a coarse certification bound; the instance-dependent refinement $\sum_i w_i \lambda_2(A_i)(1 - \|P_F^\star u_i\|^2)$ identifies the two mechanisms controlling surrogate loss (residual spectral mass and optimiser misalignment) and is tight in three regimes (rank-1 A_i, r-coverable case, common-shape).

Position in the literature. The Grassmannian $\mathrm{Gr}(r, d)$ is the standard parameter space for rank-r linear maps, with smooth Riemannian structure inherited from $O(d)$ [3]; the Ky Fan trace maximisation principle is classical [1, 2]. Principal component analysis [4] is the corresponding single-matrix instance of trace-score optimality. The condition that several covariance-like matrices share a common eigenstructure has a long history under the names *common principal components* [5, 6] and *joint diagonalisation* [7]. Reduced-order models for linear systems are commonly built from balanced Gramians [8] or by truncating in H_2/H_∞ norms [9]; sensor-placement and output-design methods optimise estimation or reconstruction quality for a single objective [10]. Robust subspace recovery [11] addresses outlier and contamination structure, complementary to the exact-optimality focus here. Within this landscape, Propositions 4 and 17 are structural consequences of the Ky Fan trace and operator-norm variational principles respectively; what is new here is to put the two existence laws side by side, prove the strict gap between them (Theorem 21), and develop the weak-glue regret comparison (Theorems 23, 24) that quantifies suboptimality when neither exact condition holds.

Outline. Section 2 gives the setup, the two scoring rules, and the notion of a common optimal rank-r projection. Section 3 proves the trace-score theorem and instantiates it in LGDS, recovering the Bayes-risk formula and the regret-angle bound. Section 4 proves the operator-norm result and gives the linearised Jacobian-response motivation. Section 5 proves the fork. Section 6 develops the weak-glue surrogate. Section 7 gives numerical illustrations. Section 8 closes with scope and discussion.

Table 1 summarises the two main results side by side.

2. Setup

State space and projector class. The state space is $\mathbb{R}^d$ with the Euclidean inner product. A *rank-r projection* ($1 \le r < d$) is an orthogonal projector P of rank r; equivalently, $P = U^\top U$ for a row-orthonormal $U \in \mathbb{R}^{r \times d}$ with $UU^\top = I_r$. The set of rank-r projectors is naturally identified with the Grassmannian $\mathrm{Gr}(r, d)$, a smooth compact Riemannian manifold

of dimension $r(d-r)$ with metric and geodesic structure inherited from $O(d)$ [3]. Two projectors with the same range are equivalent on $\mathrm{Gr}(r,d)$.

Target family and scores. A *target family* $\mathcal{T}$ indexes the tasks. Each $t \in \mathcal{T}$ induces two PSD operators on $\mathbb{R}^d$:

- a *trace-score target operator* $M_t \succeq 0$,
- an *operator-norm target operator* $K_t \succeq 0$.

The two scoring rules considered are:

$$S_t^{\mathrm{tr}}(P) := \mathrm{tr}(PM_t), \qquad S_t^{\mathrm{op}}(P) := \lambda_{\max}(PK_tP). \tag{1}$$

The trace score is the Ky Fan trace; it is a smooth (analytic) function of $P \in \mathrm{Gr}(r,d)$. The operator-norm score is the largest eigenvalue of PK_tP; it is continuous on $\mathrm{Gr}(r,d)$. Indeed, since $\lambda_{\max}$ is 1-Lipschitz in operator norm on symmetric matrices [2], for any rank-r orthogonal projectors P, Q,

$$|\lambda_{\max}(PK_tP) - \lambda_{\max}(QK_tQ)| \le \|PK_tP - QK_tQ\| \le 2\,\|K_t\|\,\|P-Q\|. \tag{2}$$

The score is smooth on the Grassmannian stratum where the leading eigenvalue of PK_tP is simple, with non-smooth points at eigenvalue crossings [2]. The operator-norm score has a linearised Jacobian-response motivation in the local setting of §4. This is a local linear-algebra setting; "intervention" is not used here in the sense of statistical causal identification or do-calculus.

Spectral subspaces.

Definition 1 (Dominant invariant subspace; leading eigenspace)**.** *For PSD $T \succeq 0$ on $\mathbb{R}^d$ with eigenvalues $\lambda_1(T) \ge \cdots \ge \lambda_d(T) \ge 0$, a* dominant r-dimensional invariant subspace *of T is any r-dimensional subspace $V \subseteq \mathbb{R}^d$ satisfying* $\mathrm{tr}(\Pi_V T) = \sum_{i=1}^r \lambda_i(T)$ *(the Ky Fan maximum). Under strict r-gap $\lambda_r(T) > \lambda_{r+1}(T)$, V is unique and equals the span of eigenvectors of $\lambda_1(T), \ldots, \lambda_r(T)$; in degenerate cases ($\lambda_r = \lambda_{r+1}$), "dominant" simply means Ky–Fan-maximising and such subspaces need not be unique. The* leading eigenspace *is $E_{\max}(T) := \ker(T - \lambda_1(T)I)$ when $T \neq 0$, and $E_{\max}(0) := \mathbb{R}^d$ by convention.*

Common optimal projection.

Definition 2 (Common optimal rank-r projection)**.** *A rank-r orthogonal projector $P^\star$ is a* common optimal rank-r projection *for the family $\mathcal{T}$ under score S if*

$$S_t(P^\star) \;=\; \max_{P:\, P^\top = P = P^2,\, \mathrm{rank} P = r} S_t(P) \qquad \textit{for every } t \in \mathcal{T}. \tag{3}$$

Throughout this paper, P ranges over rank-r orthogonal projectors (equivalently $P \in \mathrm{Gr}(r,d)$ via $P = U^\top U$, $UU^\top = I_r$); the max *is over this set.*

The central question of the paper is when a common optimal rank-r projection exists, under each scoring rule, and what structure is forced on the family if so. The two scoring rules give different answers.

3. Trace-Score Common Optimality via r-Coherence

3.1. The general PSD theorem

Definition 3 (r-coherence)**.** *A PSD family $\{M_t\}_{t\in\mathcal{T}}$ on $\mathbb{R}^d$ is* r-coherent *if there exists a rank-r subspace $V^\star \subseteq \mathbb{R}^d$ that is a dominant r-dimensional invariant subspace of every M_t (equivalently,* $\mathrm{tr}(\Pi_{V^\star} M_t) = \sum_{i=1}^r \lambda_i(M_t)$ *for every t). Under strict r-gaps ($\lambda_r(M_t) > \lambda_{r+1}(M_t)$ for all t), this is equivalent to the equality of the top-r eigenspaces of all M_t.*

Proposition 4 (Trace-score common optimality)**.** *A common optimal rank-r projection for the PSD family $\{M_t\}$ under S^{tr} exists if and only if $\{M_t\}$ is r-coherent.*

Proof. By the Ky Fan variational principle [1, 2], for each t,

$$\max_{\operatorname{rank} P = r} \operatorname{tr}(PM_t) = \sum_{i=1}^{r} \lambda_i(M_t), \tag{4}$$

attained exactly by projectors $P = \Pi_V$ with V a dominant r-dimensional invariant subspace of M_t in the sense of Definition 1. ($\Leftarrow$) If $V^\star$ is a common dominant r-dimensional invariant subspace of every M_t, then $P^\star := \Pi_{V^\star}$ attains the maximum (4) for every t. ($\Rightarrow$) Conversely, if $P^\star$ attains the maximum for every t, then $\operatorname{ran}(P^\star)$ is a dominant r-dimensional invariant subspace of every M_t, so the family is r-coherent. Under strict r-gaps the dominant invariant subspace is unique and equals the top-r eigenspace, so all M_t share the same top-r eigenspace. ■

Remark 5 (Comparison with simultaneous diagonalisation and CPC)**.** *r-coherence is closer to a* common principal subspace *condition [5, 6, 12, 13] than to a common-eigenbasis condition. It is strictly weaker than simultaneous diagonalisability: pairwise commutativity $[M_s, M_t] = 0$ implies a common eigenbasis (and hence r-coherence for every r), and joint-diagonalisation algorithms [7] target this stronger condition. r-coherence requires only a shared dominant r-dimensional invariant subspace, not a common eigenbasis: the projections of the M_t onto the orthogonal complement of $V^\star$ may have unrelated spectral structure. The CPC literature has developed likelihood-ratio tests for shared top-r eigenspaces and partial CPC models [12, 13]; the present note treats $\{M_t\}$ as exact PSD inputs and does not address statistical testing. The new content here is the side-by-side comparison with the operator-norm objective (§4) and the strict coherence/coverability separation (§5), not the characterisation Proposition 4 on its own.*

Remark 6 (Genericity)**.** *For generic $\{M_t\}$, distinct M_t have distinct top-r eigenspaces. r-coherence is therefore a non-generic algebraic condition; outside r-coherence, the trace-score optimum is target-dependent.*

Proposition 7 (Generic non-coherence: scoped, fibrewise)**.** *Fix $1 \le r < d$, $m \ge 2$, and a PSD matrix M_1 on $\mathbb{R}^d$ with simple top-r spectrum (so $\lambda_r(M_1) > \lambda_{r+1}(M_1)$ and the top-r eigenspace V_1 is unique). Within the space of $(m-1)$-tuples $(M_2, \dots, M_m)$ of PSD matrices on $\mathbb{R}^d$ such that each M_t also has simple top-r spectrum, the subset for which $\{M_1, \dots, M_m\}$ is r-coherent is contained in a finite intersection of algebraic subvarieties each of codimension $r(d-r)$ in the space of symmetric $d \times d$ matrices; in particular it is closed, nowhere dense, and has Lebesgue measure zero. Equivalently, for M_1 fixed and $(M_2, \dots, M_m)$ ranging over an open dense set of PSD tuples with strict r-gap, no common rank-r trace-optimal projection exists.*

Proof. With V_1 the unique top-r eigenspace of M_1, let $V_1^\perp = \mathbb{R}^d \ominus V_1$ and $\Pi_{V_1}, \Pi_{V_1^\perp}$ the corresponding orthogonal projectors. By Proposition 4, a rank-r projection with range V_1 is trace-optimal for M_t iff V_1 is the top-r eigenspace of M_t (under simple top-r spectrum), and V_1 is the top-r eigenspace of M_t iff V_1 is M_t-invariant with eigenvalues matching the top r of M_t. M_t-invariance of V_1 is the off-diagonal block condition $\Pi_{V_1^\perp} M_t \Pi_{V_1} = 0$, an $(d-r) \times r$ symmetric-matrix linear equation on M_t, i.e. $r(d-r) \ge 1$ independent linear constraints on the entries of M_t. For each $t \ge 2$ this defines a codimension-$r(d-r)$ linear subvariety $\mathcal{S}_t \subset \mathrm{Sym}_d$. The r-coherent locus is $\bigcap_{t=2}^{m} \mathcal{S}_t$, of codimension at least $r(d-r)$ inside Sym_d^{m-1}, hence closed, nowhere dense, and Lebesgue null. The strict-gap simple-top condition on each M_t is an open dense condition that does not affect codimension. ■

Remark 8 (Scope of the genericity claim)**.** *Proposition 7 is fibrewise: M_1 is held fixed in the strict-gap simple-top stratum, and $(M_2, \dots, M_m)$ is the variable parameter. A global statement varying all M_t at once would require handling degenerate strata where some M_t has a multiplicity at λ_r; on those strata the dominant invariant subspace is a Grassmannian fibre rather than a unique subspace, and a separate codimension count applies. We do not pursue the global version. The fibrewise version is enough to formalise the claim that r-coherence is non-generic: for any reasonable continuous prior on $(M_2, \dots, M_m)$ with M_1 fixed, the probability of a common trace-optimal projection is zero.*

Proposition 7 formalises the genericity remark: in a typical PSD family, r-coherence is not just absent, it is measure-zero absent. A small perturbation of a coherent family destroys coherence, and there is no way to restore it without imposing $r(d-r)$ algebraic constraints. This gives the trace/operator-norm fork algebraic substance. On the generic locus where r-coherence fails, no common trace-optimal projection exists; whether a common operator-norm-optimal projection exists is a separate, much weaker condition (r-coverability), which in the simple-top case holds whenever the leading eigenvectors span at most r dimensions — a condition compatible with codimension-zero sets of families. The two existence conditions sit on opposite sides of a genericity gap.

3.2. LGDS instantiation

The trace-score theorem's classical instance is the stationary linear Gaussian dynamical system (LGDS) with squared-error prediction loss ("predictive" is the appropriate adjective in this LGDS instance). Let

$$x_{t+1} = Ax_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, Q), \tag{5}$$

with A stable and $Q \succ 0$, and let $\Sigma \succ 0$ denote the stationary covariance solving $A\Sigma A^\top + Q = \Sigma$ [14]. A prediction target is a pair (C, τ) with $C \in \mathbb{R}^{p\times d}$ and horizon $\tau \geq 1$; the goal is to predict $Cx_{t+\tau}$ from a rank-r projection of the current state. The map $P = U^\top U$ is an orthogonal projection in the *whitened* coordinates $\Sigma^{-1/2}x$; in original state coordinates the observed subspace is its $\Sigma^{-1/2}$-pullback, and the observation map is $\phi_U(x) = U\Sigma^{-1/2}x$. Pre-whitening places all candidate projections on a common isotropic footing.

Lemma 9 (Bayes risk for LGDS)**.** *For projector $P = U^\top U$ with $UU^\top = I_r$, define the linear map $\phi_U(x) = U\Sigma^{-1/2}x$. At stationarity, the squared-error Bayes risk for predicting $Cx_{t+\tau}$ from $\phi_U(x_t)$ is*

$$\mathcal{R}(U;\, C, \tau) = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}(UM_{C,\tau}U^\top), \tag{6}$$

where the target information matrix *is*

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top CA^\tau\Sigma^{1/2}. \tag{7}$$

The matrix $M_{C,\tau}$ is symmetric PSD because $\Sigma^{1/2}$ is symmetric and $M_{C,\tau} = (CA^\tau\Sigma^{1/2})^\top(CA^\tau\Sigma^{1/2})$.

Proof. $x_{t+\tau} = A^\tau x_t + w_\tau$ with w_τ independent of x_t, so $\mathbb{E}[x_{t+\tau}x_t^\top] = A^\tau\Sigma$. With $z = U\Sigma^{-1/2}x_t$, $\mathrm{Cov}(z, z^\top) = U\Sigma^{-1/2}\Sigma\Sigma^{-1/2}U^\top = I_r$ and $\mathrm{Cov}(Cx_{t+\tau}, z^\top) = CA^\tau\Sigma \cdot \Sigma^{-1/2}U^\top = CA^\tau\Sigma^{1/2}U^\top$. For jointly Gaussian $(Cx_{t+\tau}, z)$ with $\mathrm{Cov}(z, z^\top) = I_r$ the conditional variance is $\mathrm{Var}(Cx_{t+\tau}\,|\,z) = \mathrm{Var}(Cx_{t+\tau}) - \mathrm{Cov}(Cx_{t+\tau}, z^\top)\mathrm{Cov}(z, Cx_{t+\tau})$, and the trace of this conditional variance is the MMSE $\mathcal{R}(U; C, \tau)$. Substituting the covariances gives $\mathcal{R} = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}(U\Sigma^{1/2}(A^\tau)^\top C^\top CA^\tau\Sigma^{1/2}U^\top)$, which is (6). ■

Since $\mathrm{tr}(C\Sigma C^\top)$ is independent of U, minimising $\mathcal{R}(U; C, \tau)$ is equivalent to maximising $\mathrm{tr}(UM_{C,\tau}U^\top) = \mathrm{tr}(PM_{C,\tau})$ where $P = U^\top U$. Proposition 4 applied to the LGDS family $\{M_{C_k,\tau_k}\}$ gives:

Corollary 10 (LGDS prediction iff)**.** *A common rank-r projection minimising squared-error prediction risk for every $(C_k, \tau_k) \in \mathcal{T}$ exists if and only if $\{M_{C_k,\tau_k}\}$ is r-coherent.*

Lemma 11 (PSD families are realisable as LGDS prediction tasks)**.** *For any finite family $\{M_t\}_{t=1}^m$ of PSD matrices on $\mathbb{R}^d$, there exist a stable LGDS (A, Q) and output matrices $\{C_t\}_{t=1}^m$ at horizon $\tau = 1$ such that*

$$M_{C_t,1} = \Sigma^{1/2}A^\top C_t^\top C_t A\,\Sigma^{1/2} = M_t \quad \textit{for every } t.$$

Proof. Pick any $a \in (0, 1)$ and set $A = aI_d$, $Q = (1 - a^2)I_d$; then $A\Sigma A^\top + Q = \Sigma$ is solved by $\Sigma = I_d$, so $\Sigma^{1/2} = I_d$. For each t, let $C_t \in \mathbb{R}^{d\times d}$ be any matrix with $C_t^\top C_t = a^{-2}M_t$ (e.g. the symmetric square root of $a^{-2}M_t$). Then $M_{C_t,1} = A^\top C_t^\top C_t A = a^2(a^{-2}M_t) = M_t$. ■

Lemma 11 converts every PSD example used elsewhere in the paper (in particular the fork witness of §5) into a concrete LGDS prediction-task family with the same target information matrices. The fork is therefore not an abstract PSD curiosity: it is an explicit 3×3 LGDS prediction example. This is a realisability statement only — it shows that the PSD examples used in this paper can be exhibited inside the LGDS prediction-task framework — and is not a claim that arbitrary real-world dynamical systems generate arbitrary PSD families of target information matrices in practice.

3.3. Task regret and the angle bound

The deviation of P from $P^\star$ admits a sharp two-sided characterisation in principal angles [15]. Let M be PSD with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d$ and top-r eigenspace projector $P^\star$. Define the *task regret*

$$\mathrm{Reg}(P;\, M) := \mathrm{tr}(P^\star M) - \mathrm{tr}(PM) = \sum_{i=1}^{r} \lambda_i(M) - \mathrm{tr}(PM) \;\geq\; 0. \tag{8}$$

Corollary 12 (Regret–angle bound)**.** *Let* $\|\sin\Theta\|_F^2 := \|\sin\Theta(P, P^\star)\|_F^2$ *be the squared Frobenius norm of the principal-angle sines between* $\mathrm{ran}(P)$ *and* $\mathrm{ran}(P^\star)$ *[15, 16, 17]. Then*

$$(\lambda_r - \lambda_{r+1})\|\sin\Theta\|_F^2 \;\le\; \mathrm{Reg}(P;\, M) \;\le\; (\lambda_1 - \lambda_d)\|\sin\Theta\|_F^2. \tag{9}$$

Proof. Let v_i be unit eigenvectors of M with eigenvalues λ_i, and u_k orthonormal vectors spanning $\mathrm{ran}(P)$. Set $p_i := \sum_{k=1}^r (v_i^\top u_k)^2$. Then $\mathrm{tr}(PM) = \sum_i \lambda_i p_i$ and $\sum_i p_i = r$, while $\|\sin\Theta\|_F^2 = \sum_{i\le r}(1-p_i) = \sum_{i>r} p_i$. Centering at λ_{r+1}:

$$\mathrm{Reg}(P;\, M) = \sum_{i\le r}(\lambda_i - \lambda_{r+1})(1-p_i) + \sum_{i>r}(\lambda_{r+1} - \lambda_i)\, p_i.$$

Both sums are nonnegative. The lower bound follows by dropping the second sum and bounding $\lambda_i - \lambda_{r+1} \ge \lambda_r - \lambda_{r+1}$. The upper bound follows by bounding the two coefficients by $\lambda_1 - \lambda_{r+1}$ and $\lambda_{r+1} - \lambda_d$ respectively. ■

Corollary 13 (Approximate r-coherence)**.** *For a finite trace-score family* $\{M_a\}$*, define the* minimax trace-score regret

$$\Gamma_{\mathrm{tr}} := \min_{P\in\mathrm{Gr}(r,d)} \max_a \mathrm{Reg}(P;\, M_a). \tag{10}$$

$\Gamma_{\mathrm{tr}} = 0$ *iff* $\{M_a\}$ *is r-coherent. If there exists* $P \in \mathrm{Gr}(r,d)$ *with* $\|\sin\Theta(P, P_a^\star)\|_F^2 \le \varepsilon^2$ *for every a, then* $\Gamma_{\mathrm{tr}} \le \Lambda\,\varepsilon^2$*, where* $\Lambda := \max_a[\lambda_1(M_a) - \lambda_d(M_a)]$.

Proof. The first claim is Proposition 4 applied to the family. The second applies the upper bound of Corollary 12 uniformly in a. ■

Remark 14 (Contrapositive certificate)**.** *The contrapositive of the upper bound gives a one-sided practical certificate: if* $\Gamma_{\mathrm{tr}} > \delta$ *for some* $\delta > 0$*, then no rank-r projection lies within squared principal-angle distance* δ/Λ *of every task-optimal subspace* $P_a^\star$*. This is not a matching lower bound (the upper bound need not be tight), but rules out approximate r-coherence at the corresponding scale.*

4. Operator-Norm Common Optimality via *r*-Coverability

4.1. Motivation for the operator-norm score

The operator-norm score $\lambda_{\max}(PK_tP)$ is not a Ky Fan trace. A natural setting is local linearised response: at state x, an input perturbation $\delta u \in \mathbb{R}^q$ induces $\delta y = J_x\delta u$ via the local Jacobian J_x, and a rank-r projection P restricts which output directions are visible. The largest squared response visible to P over unit-norm inputs is

$$\max_{\|\delta u\|=1} \|PJ_x\delta u\|^2 = \lambda_{\max}(PJ_xJ_x^\top P) = \lambda_{\max}(PK_xP), \qquad K_x := J_xJ_x^\top. \tag{11}$$

Hence $\lambda_{\max}(PK_tP)$ measures the *strongest* accessible squared response under P, with $K_t \ge 0$ a local PSD response operator; a trace alternative $\mathrm{tr}(PK_t)$ measures *total* accessible squared response and answers a different question. The operator-norm choice is appropriate when the task is to detect or recover the strongest linear effect rather than aggregate sensitivity.

4.2. Local maximiser structure

Proposition 15 (Fixed-target operator-norm maximiser)**.** *For fixed PSD* $K \ge 0$ *on* $\mathbb{R}^d$ *with leading invariant subspace* $E_{\max}(K)$ *(the eigenspace of* $\lambda_1(K)$*),*

$$\arg\max_{\mathrm{rank}P=r} \lambda_{\max}(PKP) = \{P : \mathrm{rank}(P) = r,\ \mathrm{ran}(P) \cap E_{\max}(K) \ne \{0\}\}, \tag{12}$$

and the maximum value is $\lambda_1(K)$*.*

Proof. For any unit $v \in \mathbb{R}^d$, $v^\top Kv \le \lambda_1(K)$, with equality iff $v \in E_{\max}(K)$. For rank-r P, $\lambda_{\max}(PKP) = \sup_{v\in\mathrm{ran}(P),\,\|v\|=1} v^\top Kv$. This supremum equals $\lambda_1(K)$ iff $\mathrm{ran}(P)$ contains a unit vector in $E_{\max}(K)$, i.e., $\mathrm{ran}(P) \cap E_{\max}(K) \ne \{0\}$. ■

The contrast with the trace-score case is sharp: under $S^{\mathrm{tr}} = \mathrm{tr}(PM)$ the optimal P must have its range *equal to* a dominant r-dimensional invariant subspace of M; under $S^{\mathrm{op}} = \lambda_{\max}(PKP)$ it suffices that $\mathrm{ran}(P)$ *intersects* the leading invariant subspace of K nontrivially. The local problem is *rank-1-driven* in the sense that the maximiser is determined by a single direction in $E_{\max}(K)$, regardless of r: any unit vector lying in $E_{\max}(K) \cap \mathrm{ran}(P)$ already realises the maximum value $\lambda_1(K)$.

4.3. r-coverability and the operator-norm theorem

Definition 16 (r-coverability)**.** *A PSD family $\{K_t\}$ is r-*coverable *if there exists a rank-r subspace $V \subseteq \mathbb{R}^d$ with $V \cap E_{\max}(K_t) \neq \{0\}$ for every t. When each K_t has a simple top eigenvalue with top eigenvector u_t, r-coverability reduces to* $\dim \operatorname{span}\{u_t\} \leq r$.

Proposition 17 (Operator-norm common optimality)**.** *A common optimal rank-r projection for the PSD family $\{K_t\}$ under S^{op} exists if and only if $\{K_t\}$ is r-coverable.*

Proof. By Proposition 15, $P^\star$ is a common optimal rank-r projection under S^{op} iff $\operatorname{ran}(P^\star) \cap E_{\max}(K_t) \neq \{0\}$ for every t. Existence of such a rank-r subspace is the definition of r-coverability. ■

Remark 18 (Coverability is weaker than coherence)**.** *r-coherence demands that one rank-r subspace* equal *a dominant r-dimensional invariant subspace of every M_t. r-coverability demands only that the rank-r subspace* hit *each $E_{\max}(K_t)$.*

Remark 19 (Testing r-coverability)**.** *r-coverability is a finite-dimensional subspace-transversal / incidence problem on the Grassmannian: $V \in \mathrm{Gr}(r, d)$ must satisfy a Schubert-type incidence condition $V \cap E_{\max}(K_t) \neq \{0\}$ for every t. For the present note we use the following simple sufficient and necessary checks, which already cover the cases of interest.*

- Simple-top reduction. *If $\dim E_{\max}(K_t) = 1$ with top eigenvector u_t for every t, V must contain each u_t, so r-coverability is equivalent to* $\dim \operatorname{span}\{u_t\} \leq r$.
- Sufficient condition (any multiplicity). *If $\dim \sum_t E_{\max}(K_t) \leq r$, any r-dimensional subspace containing this sum is an r-cover.*

The non-simple case becomes a Schubert-incidence feasibility problem on $\mathrm{Gr}(r, d)$; a full algorithmic complexity classification is outside the scope of this note. The simple-top reduction is sufficient but in general strictly stronger than r-coverability, since freedom to choose unit $u_t \in E_{\max}(K_t)$ may bring $\dim \operatorname{span}\{u_t\}$ below r when some $E_{\max}(K_t)$ are multidimensional. A concrete witness ($d = 3$, $r = 1$): let $E_1 = \operatorname{span}\{e_1, e_2\}$ and $E_2 = \operatorname{span}\{e_2, e_3\}$. The 1-plane $V = \operatorname{span}\{e_2\}$ intersects both, so 1-coverability holds. However, the simple-top span test applied to the choice $u_1 = e_1 \in E_1$, $u_2 = e_3 \in E_2$ gives $\dim \operatorname{span}\{u_1, u_2\} = 2 > 1$ and would incorrectly reject coverability. This shows that the multidimensional case is genuinely a subspace-hitting problem, not a span dimension count on representatives.

4.4. Sufficient condition for the simple-top case

Corollary 20 (Simple-top r-coverability)**.** *Suppose $\{K_t\}_{t \in T}$ is indexed by a finite-support measure $\mu = \sum_i w_i \delta_{x_i}$ on a state space, with $K_i := K_{x_i}$ having simple top eigenvalues and top eigenvectors u_i. Then $\{K_i\}$ is r-coverable iff* $\dim \operatorname{span}\{u_i\} \leq r$. *In that case any $P^\star$ whose range contains* $\operatorname{span}\{u_i\}$ *achieves the global operator-norm optimum $S_i^{\mathrm{op}}(P^\star) = \lambda_1(K_i)$ for every i. This regime is informally called* strong glue.

Proof. Direct from Definition 16 and Proposition 15. ■

5. The Fork: Coherence Implies Coverability Strictly

The two structural conditions are not interchangeable. We write a single PSD family as $\{T_t\}$ in this section, isolating the algebraic relation between r-coherence and r-coverability from the trace-score (M_t) vs. operator-norm (K_t) interpretations of §3–4.

Theorem 21 (Coherence implies coverability strictly)**.** *For a single PSD family $\{T_t\}$ on $\mathbb{R}^d$:*

(i) r-coherence implies r-coverability.
(ii) The converse fails in general.

Proof. (i) Let $\{T_t\}$ be r-coherent with shared dominant subspace $V^\star$, and fix t. A Ky–Fan-optimal $V^\star$ contains a top eigenvector of T_t: diagonalising $\Pi_{V^\star} T_t \Pi_{V^\star}$ on $V^\star$, the eigenvalues $\mu_1 \geq \cdots \geq \mu_r$ satisfy $\mu_k \leq \lambda_k(T_t)$ by Cauchy interlacing [2] and $\sum_k \mu_k = \sum_i \lambda_i(T_t)$, forcing $\mu_1 = \lambda_1(T_t)$. Since $v_1^\top T_t v_1 = \mu_1 \|v_1\|^2 = \lambda_1(T_t)\|v_1\|^2$, the Rayleigh quotient equality case forces $v_1 \in E_{\max}(T_t)$. Therefore $V^\star \cap E_{\max}(T_t) \neq \{0\}$ for every t, so $V^\star$ is an r-cover.

(ii) For the converse, take $d = 3$, $r = 2$ and

$$T_1 = \mathrm{diag}(5, 4, 0), \qquad T_2 = \mathrm{diag}(5, 0, 4). \tag{13}$$

Both have simple top eigenvalue 5 with top eigenvector e_1, so any rank-2 subspace containing e_1 is a 2-cover; the family is 2-coverable. However, the top-2 eigenspaces are $\mathrm{Top}_2(T_1) = \mathrm{span}\{e_1, e_2\}$ and $\mathrm{Top}_2(T_2) = \mathrm{span}\{e_1, e_3\}$, which are distinct. Their intersection is $\mathrm{span}\{e_1\}$, of dimension $1 < r = 2$, so no rank-2 subspace is contained in both top-2 eigenspaces; the family is not 2-coherent. ■

The witness isolates the core insight: the same leading vector e_1 is enough for operator-norm optimality (which requires only nontrivial intersection with $E_{\max}$), but not for trace optimality, which also requires capturing the second dominant direction — and the second dominant directions of T_1 and T_2 disagree.

Non-diagonal version of the witness. The diagonal form is not essential. For any $Q \in O(3)$, set

$$\widetilde{T}_1 = Q^\top \mathrm{diag}(5, 4, 0) Q, \qquad \widetilde{T}_2 = Q^\top \mathrm{diag}(5, 0, 4) Q.$$

Then $\widetilde{T}_1, \widetilde{T}_2$ are generally non-diagonal in the standard basis but have the same coherence/coverability behaviour: both share the leading unit vector $Q^\top e_1$, so they are 2-coverable, while their dominant 2-dimensional subspaces are $Q^\top \mathrm{span}\{e_1, e_2\}$ and $Q^\top \mathrm{span}\{e_1, e_3\}$, whose intersection is one-dimensional. Thus the fork is basis-invariant and not an artefact of simultaneous diagonality.

Remark 22 (Trace vs operator-norm dissociation)**.** *Theorem 21 establishes an algebraic gap between coherence and coverability for a single PSD family. In a setting where the trace and operator-norm operators $\{M_t\}$, $\{K_t\}$ are different functions of the same task index, the two existence conditions decouple. A common optimal projection under S^{op} can exist while no common optimal projection under S^{tr} does, and vice versa: the existence law for a common optimal projection depends on the scoring rule, not just the family.*

6. Surrogate Approximation When Coverability Fails

When the operator-norm family $\{K_i\}$ has $\dim \mathrm{span}\{u_i\} > r$ (i.e. fails r-coverability), no projector achieves the global operator-norm optimum simultaneously. The relevant question becomes *approximate* common optimality under the operator-norm score: how close to optimal can a single rank-r projection come across the family, and is there a tractable surrogate for the non-Ky–Fan operator-norm objective? (The simple-top r-coverable case and the non-coverable surrogate regime are sometimes called "strong glue" and "weak glue" respectively; these are informal shorthands, not additional assumptions.) The bounds below control *value regret* $F(P_F^\star) - F(P_B^\star)$, not subspace distance: as the $d = 3$ counterexample of §7 shows, $P_F^\star$ and $P_B^\star$ can be geometrically separated even when the value gap is small.

6.1. Setup and surrogate

Standing assumption for this section. The regime is finite-support, $\mu = \sum_{i=1}^N w_i \delta_{x_i}$ with $w_i > 0$, $\sum w_i = 1$, and each local Gram $A_i := J_i J_i^\top \succeq 0$ has *simple* top eigenvalue ($\lambda_1(A_i) > \lambda_2(A_i) \geq 0$), with top eigenvector u_i and residual ratio $\kappa_i := \lambda_2(A_i)/\lambda_1(A_i) \in [0, 1)$. Repeated leading eigenspaces would require a projector-valued surrogate in place of $u_i u_i^\top$ and are outside the scope of this section.

The regime-averaged operator-norm score and its surrogate are:

$$F(P) := \sum_{i=1}^N w_i \, \lambda_{\max}(P A_i P) = \sum_{i=1}^N w_i \, S_i^{\mathrm{op}}(P), \tag{14}$$

$$\tilde{F}(P) := \mathrm{tr}(P B_\lambda), \qquad B_\lambda := \sum_{i=1}^N w_i \, \lambda_1(A_i) \, u_i u_i^\top. \tag{15}$$

Thus F is the μ-weighted average of the per-target operator-norm score $S_i^{\mathrm{op}}(P)$ from (1) with $K_i := A_i$. The surrogate $\tilde{F}$ is a Ky Fan trace and is therefore maximised by any rank-r projection whose range is a dominant invariant subspace of B_λ.

The choice of B_λ is not arbitrary. For each i, the lower bound (19) gives $\lambda_1(A_i)\|Pu_i\|^2 = \mathrm{tr}(P\,\lambda_1(A_i)u_iu_i^\top) \le \lambda_{\max}(PA_iP)$; summing the rank-one lower contributions with weights w_i yields exactly $\mathrm{tr}(PB_\lambda) = \tilde{F}(P)$. B_λ is the Ky Fan matrix collecting the rank-one leading-direction lower approximation to $F(P)$. It is exact when every A_i is rank-1 (Theorem 23) and is controlled by the residual spectra $\lambda_2(A_i)$ when not. Let $P_F^\star \in \arg\max_{\mathrm{rank}P=r} F(P)$ (a true operator-norm optimiser) and $P_B^\star \in \arg\max_{\mathrm{rank}P=r} \tilde{F}(P)$ (a surrogate optimiser); under strict r-gap of B_λ both are unique up to range equivalence on $\mathrm{Gr}(r,d)$, while under degeneracy both are interpreted as *any* element of the corresponding argmax.

6.2. *The sandwich and coarse regret*

Theorem 23 (Weak-glue sandwich and regret)**.** *For all rank-r projectors P:*

$$\tilde{F}(P) \;\le\; F(P) \;\le\; \tilde{F}(P) + \sum_{i=1}^{N} w_i\,\lambda_2(A_i). \tag{16}$$

Consequently,

$$F(P_F^\star) - F(P_B^\star) \;\le\; \sum_{i=1}^{N} w_i\,\lambda_2(A_i). \tag{17}$$

$\tilde{F} = F$ pointwise iff every A_i is rank-1 (i.e. $\kappa_i = 0$ for all i).

Proof. Per-term bound. Fix i and let $A_i = \sum_{k=1}^{d} \lambda_k(A_i)e_ke_k^\top$ be the spectral decomposition with $e_1 = u_i$. For any unit $v \in \mathrm{ran}(P)$,

$$v^\top A_i v = \sum_k \lambda_k(A_i)(e_k^\top v)^2 \le \lambda_1(A_i)(u_i^\top v)^2 + \lambda_2(A_i)\sum_{k\ge 2}(e_k^\top v)^2 = \lambda_1(A_i)(u_i^\top v)^2 + \lambda_2(A_i)(1-(u_i^\top v)^2).$$

Since $v \in \mathrm{ran}(P)$ and $P^\top = P$, $u_i^\top v = u_i^\top Pv = (Pu_i)^\top v$, so by Cauchy–Schwarz $(u_i^\top v)^2 \le \|Pu_i\|^2\|v\|^2 = \|Pu_i\|^2$. The map $t \mapsto \lambda_1(A_i)t + \lambda_2(A_i)(1-t)$ is non-decreasing on $[0,1]$ since $\lambda_1 \ge \lambda_2$, so $v^\top A_i v \le \lambda_1(A_i)\|Pu_i\|^2 + \lambda_2(A_i)(1-\|Pu_i\|^2)$. Taking the supremum over unit $v \in \mathrm{ran}(P)$ gives the upper bound

$$\lambda_{\max}(PA_iP) \le \lambda_1(A_i)\|Pu_i\|^2 + \lambda_2(A_i)(1-\|Pu_i\|^2). \tag{18}$$

For the lower bound, when $\|Pu_i\| > 0$ test $v = Pu_i/\|Pu_i\| \in \mathrm{ran}(P)$: $v^\top A_i v = (Pu_i)^\top A_i(Pu_i)/\|Pu_i\|^2 \ge \lambda_1(A_i)(u_i^\top Pu_i)^2/\|Pu_i\|^2 = \lambda_1(A_i)\|Pu_i\|^2$, using the orthogonal-projector identity $u_i^\top Pu_i = u_i^\top P^2u_i = \|Pu_i\|^2$ and dropping the nonnegative $\lambda_k \ge 2$ terms. When $\|Pu_i\| = 0$ the bound is trivial. Thus

$$\lambda_{\max}(PA_iP) \ge \lambda_1(A_i)\|Pu_i\|^2. \tag{19}$$

Sandwich. Summing (19) weighted by w_i gives $\tilde{F}(P) \le F(P)$; summing (18) and using $1-\|Pu_i\|^2 \le 1$ gives $F(P) \le \tilde{F}(P) + \sum_i w_i\lambda_2(A_i)$.

Coarse regret (17). Decompose $F(P_F^\star) - F(P_B^\star) = [F(P_F^\star) - \tilde{F}(P_F^\star)] + [\tilde{F}(P_F^\star) - \tilde{F}(P_B^\star)] + [\tilde{F}(P_B^\star) - F(P_B^\star)]$. The middle term is ≤ 0 by surrogate optimality of $P_B^\star$; the third is ≤ 0 by the lower sandwich; the first is bounded by $\sum_i w_i\lambda_2(A_i)$ from (18) summed.

Pointwise equality $\tilde{F} = F$. For each i, define the per-term gap $\delta_i(P) := \lambda_{\max}(PA_iP) - \lambda_1(A_i)\|Pu_i\|^2 \ge 0$ by (19). $F(P) - \tilde{F}(P) = \sum_i w_i\delta_i(P)$ with positive weights and nonnegative summands, so $F = \tilde{F}$ pointwise iff $\delta_i(P) = 0$ for every i and every rank-r projector P. Fix i and any $k \ne 1$ with $\lambda_k(A_i) > 0$ (if no such k exists, A_i is rank-1). Since $r < d$ and $\{e_2,\dots,e_d\} \perp u_i$ (the eigenvectors of A_i), the orthogonal complement of u_i has dimension $d-1 \ge r$, so we may choose a rank-r subspace V that contains e_k and is orthogonal to u_i. For $P = \Pi_V$: $\|Pu_i\| = 0$, hence $\delta_i(P) = \lambda_{\max}(PA_iP) \ge e_k^\top A_ie_k = \lambda_k(A_i) > 0$, contradicting $\delta_i \equiv 0$. Therefore $\lambda_k(A_i) = 0$ for all $k \ne 1$, i.e. A_i is rank-1. Conversely, if every A_i is rank-1, $A_i = \lambda_1(A_i)u_iu_i^\top$, and $\lambda_{\max}(PA_iP) = \lambda_1(A_i)\|Pu_i\|^2$ exactly, so $F = \tilde{F}$. ■

6.3. Instance-dependent regret

The coarse bound (17) treats every state's residual spectral mass as fully adversarial. A tighter bound accounts for the alignment of $P_F^\star$ with each local peak direction u_i.

Theorem 24 (Instance-dependent regret)**.**

$$F(P_F^\star) - F(P_B^\star) \;\le\; \sum_{i=1}^{N} w_i\, \lambda_2(A_i)(1 - \|P_F^\star u_i\|^2). \tag{20}$$

Proof. By the three-term decomposition in the proof of Theorem 23, $F(P_F^\star) - F(P_B^\star) \le F(P_F^\star) - \tilde{F}(P_F^\star)$. Applying the per-term upper bound (18) at $P = P_F^\star$ for every i and summing with weights w_i:

$$F(P_F^\star) - \tilde{F}(P_F^\star) = \sum_i w_i[\lambda_{\max}(P_F^\star A_i P_F^\star) - \lambda_1(A_i)\|P_F^\star u_i\|^2] \le \sum_i w_i \lambda_2(A_i)(1 - \|P_F^\star u_i\|^2).$$

■

Remark 25 (Two mechanisms)**.** *The instance-dependent bound factors regret at state i as the product of two independent mechanisms: (a) residual spectral mass $\lambda_2(A_i) > 0$, meaning J_i carries energy beyond its top singular direction; (b) optimizer misalignment $1 - \|P_F^\star u_i\|^2 > 0$, meaning the true operator-norm optimum fails to capture the local peak direction. Either mechanism alone produces no regret: rank-1 Jacobians contribute nothing regardless of alignment, and perfectly aligned states contribute nothing regardless of $\lambda_2(A_i)$.*

Remark 26 (Why the bounds are conservative)**.** *The coarse bound (17) is loose because it treats all residual spectral mass $\lambda_2(A_i)$ as fully accessible to the optimiser and as fully adversarial in every direction. The instance-dependent bound (20) removes the worst-case step $1-\|Pu_i\|^2 \le 1$, but remains conservative because the per-term upper bound (18) still replaces all subleading eigenvalues of A_i by $\lambda_2(A_i)$ in every orthogonal direction, ignoring spectral decay below $\lambda_2(A_i)$. Both bounds are certification bounds, not tight estimators of the value gap.*

Remark 27 (Why this looseness is unavoidable in the present information)**.** *The looseness is intrinsic to a bound stated only in terms of $\lambda_1(A_i), \lambda_2(A_i), u_i$, and $\|Pu_i\|^2$. Such a bound discards the orientation and decay of the subleading spectrum within $u_i^\perp$. Two families with the same $\lambda_1, \lambda_2, u_i$ and the same $\|Pu_i\|^2$ can have different values of $\lambda_{\max}(PA_iP)$, depending on whether the projection aligns with the λ_2-eigenvector or with lower-energy residual directions. The estimates here should therefore be read as worst-case certificates under limited spectral information. A sharper estimate would require retaining at least the projected residual matrix $P(I - u_iu_i^\top)A_i(I - u_iu_i^\top)P$, which defeats part of the purpose of the rank-one surrogate.*

6.4. Equality conditions

Corollary 28 (Three exact regimes)**.** *$F(P_F^\star) - F(P_B^\star) = 0$, i.e. the surrogate optimum equals the true optimum, in each of the following cases:*

(R1) Rank-1 Jacobians: *$\lambda_2(A_i) = 0$ for every i. Then $\tilde{F} = F$ pointwise.*

(R2) r-coverable case (informally, "strong glue"): $\dim \operatorname{span}\{u_i\} \le r$. *Both $P_F^\star$ and $P_B^\star$ achieve the global maximum $\sum_i w_i\lambda_1(A_i)$.*

(R3) Common-shape family: *$A_i = \alpha_i A$ for some fixed $A \succeq 0$ and scalars $\alpha_i > 0$. Then F and $\tilde{F}$ have identical maximiser sets, namely $\{P : \operatorname{rank}(P) = r,\ \operatorname{ran}(P) \cap E_{\max}(A) \neq \{0\}\}$.*

Proof. (R1) is Theorem 23. (R2): r-coverability gives $\|P_B^\star u_i\| = 1$ for every i since $P_B^\star$ contains $\operatorname{span}\{u_i\}$, so the lower sandwich gives $F(P_B^\star) \ge \sum_i w_i\lambda_1(A_i)$. The trivial upper bound is also $\sum_i w_i\lambda_1(A_i)$, so $P_B^\star$ achieves the global maximum. (R3): $F(P) = (\sum_i w_i\alpha_i)\lambda_{\max}(PAP)$ and $B_\lambda = (\sum_i w_i\alpha_i)\lambda_1(A)\,\Pi_{E_{\max}(A)}$, so both objectives are proportional to the operator-norm criterion of the single matrix A and have the same maximiser set $\{P : \operatorname{ran}(P) \cap E_{\max}(A) \neq \{0\}\}$ by Proposition 15. ■

In all three regimes the instance-dependent bound vanishes: either $\lambda_2(A_i) = 0$, or $\|P_F^\star u_i\|^2 = 1$ for every i. Outside these regimes the surrogate optimum need not coincide with the true optimum, but Theorem 24 controls the gap.

Remark 29 (Value regret vs subspace distance)**.** *Theorems 23–24 control* value regret $F(P_F^\star) - F(P_B^\star)$, *not subspace distance $\|\sin\Theta(P_F^\star, P_B^\star)\|_F$: $P_B^\star$ can be geometrically separated from $P_F^\star$ while the value gap remains small (the $d = 3$, $r = 1$ counterexample of Section 7 gives $\approx 11.8^\circ$ separation at 1.5% value gap).*

Table 2: Trace-score regret matrix, Condition A ($d = 8$, $r = 2$). Values from the exact Bayes risk formula (6).

Projection	Reg on T_1	Reg on T_2
$P_1^\star$ (optimal for T_1)	0.000	0.036
$P_2^\star$ (optimal for T_2)	0.763	0.000
P_{joint} (minimax)	0.026	0.026

7. Numerical Illustrations

This section gives three illustrations. The first instantiates the trace-score theorem in stationary LGDS. The second is the fork witness from Theorem 21. The third is a 3×3 weak-glue example showing the gap between the coarse and instance-dependent regret bounds. All numerical values are computed exactly: closed-form Bayes risk in the LGDS case, exact eigendecomposition for the small-dimensional examples. There is no Monte Carlo simulation.

7.1. Trace score: LGDS instance

Setup. State dimension $d = 8$, projection rank $r = 2$. $A = VDV^{-1}$ with eigenvalues drawn from $U[0.3, 0.85]$ and random non-orthogonal V (seed 100); $Q = 0.1\, I_8$. Σ is computed from the discrete Lyapunov equation; $\Sigma > 0$ to machine precision. The single-target optima $P_1^\star, P_2^\star$ are top-2 eigenspaces of M_1, M_2 in closed form (Proposition 4); no search is required. The minimax projection P_{joint} is reported here as an illustration of $\Gamma_{\text{tr}} > 0$ only; the qualitative existence statement does not depend on numerical optimisation.

Condition A: target-dependent. Targets $T_1 = ([e_1^\top; e_2^\top], 1)$ and $T_2 = ([e_5^\top; e_6^\top], 5)$. With non-normal A, A^1 and A^5 rotate state-space directions differently; the principal angles between the top-2 eigenspaces of M_1 and M_2 are 54.1° and 24.0°, confirming that no single rank-2 projection is simultaneously trace-score-optimal. By Corollary 13, the minimax trace-score regret $\Gamma_{\text{tr}} > 0$ certifies non-coherence; numerically, $\Gamma_{\text{tr}} = 0.026 > 0$. Table 2 gives the regret structure.

Condition B: structurally coherent. A target family is constructed by design to be 2-coherent: normal $A_{\text{sym}} = VDV^\top$ with orthogonal V, common C whose rows are top-2 eigenvectors of A, varying horizon $\tau \in \{1, 2, 3, 5\}$. The four resulting $M_{C,\tau}$ share the same dominant 2-dimensional eigenspace by construction, and the four optimal projections are identical: maximum pairwise principal angle is 0.000° to machine precision. Under $\varepsilon = 0.1$ perturbations to C (20 random draws), the maximum principal angle is 0.102 rad, well below the $1.5\varepsilon = 0.15$ rad threshold, illustrating that approximate coherence is quantified via principal angles (Corollary 13).

7.2. Fork: operator-norm common optimum without trace-score common optimum

The fork witness from (13): $d = 3$, $r = 2$, $T_1 = \text{diag}(5, 4, 0)$, $T_2 = \text{diag}(5, 0, 4)$.

Operator-norm score (assigning $K_t = T_t$). Both K_t have simple top eigenvalue 5 with top eigenvector e_1. Thus $\dim \text{span}\{u_t\} = 1 \le 2 = r$, so $\{K_t\}$ is 2-coverable. Any rank-2 projection whose range contains e_1—e.g. $P_{\text{cov}} = \Pi_{\text{span}\{e_1, v\}}$ for any unit $v \perp e_1$—satisfies $S_t^{\text{op}}(P_{\text{cov}}) = \lambda_1(K_t) = 5$ for both t. A common optimal rank-2 projection under S^{op} exists.

Trace score (assigning $M_t = T_t$). $\text{Top}_2(M_1) = \text{span}\{e_1, e_2\}$ and $\text{Top}_2(M_2) = \text{span}\{e_1, e_3\}$. The intersection is $\text{span}\{e_1\}$, of dimension $1 < r$. Hence no rank-2 subspace is contained in both top-2 eigenspaces; $\{M_t\}$ is not 2-coherent, so no common optimal rank-2 projection under S^{tr} exists. The minimax trace-score regret is

$$\Gamma_{\text{tr}}^{\text{fork}} = \min_{P \in \text{Gr}(2,3)} \max_{t \in \{1,2\}} \left[\lambda_1(M_t) + \lambda_2(M_t) - \text{tr}(PM_t)\right] = 2. \tag{21}$$

Derivation. Any $P \in \text{Gr}(2, 3)$ that is not a 2-cover certifies $\Gamma > 0$ by Proposition 15, so the minimum is attained at a 2-cover, i.e. a P with $e_1 \in \text{ran}(P)$. Such a P has $\text{ran}(P) = \text{span}\{e_1, v\}$ for a unit $v \in \text{span}\{e_2, e_3\}$. Write $v = \alpha e_2 + \beta e_3$ with $\alpha^2 + \beta^2 = 1$. Then $\text{tr}(PM_1) = \lambda_1 + 4\alpha^2 = 5 + 4\alpha^2$ and $\text{tr}(PM_2) = 5 + 4\beta^2$, so the two regrets are $4(1 - \alpha^2) = 4\beta^2$ and

$4(1-\beta^2) = 4\alpha^2$. The max of these two is minimised when $\alpha^2 = \beta^2 = \frac{1}{2}$, giving $\Gamma_{\mathrm{tr}}^{\mathrm{fork}} = 2$, attained by $v = (e_2 \pm e_3)/\sqrt{2}$ and the corresponding rank-2 projection $P = \Pi_{\mathrm{span}\{e_1,(e_2+e_3)/\sqrt{2}\}}$.
This gives an exact quantitative separation: in the same family for which the operator-norm common optimum is exact, the best possible common trace-score projection incurs worst-case regret 2. Thus the fork is not merely existential; it has a finite minimax gap under the trace objective.

7.3. Weak-glue counterexample

A two-state regime in $\mathbb{R}^3$ illustrates that the coarse regret bound $\sum_i w_i \lambda_2(A_i)$ can grossly overestimate the actual regret, while still detecting the qualitative point that the surrogate optimiser need not equal the true optimiser.

Construction. $N = 2$, $d = 3$, $r = 1$, equal weights $w_1 = w_2 = \frac{1}{2}$, with

$$A_1 = \mathrm{diag}(3, 2, 0), \qquad A_2 = \begin{pmatrix} 0.9 & 0.6 & 0 \\ 0.6 & 0.9 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{22}$$

A_1: $u_1 = e_1$, $\lambda_1 = 3$, $\lambda_2 = 2$, $\kappa_1 = 2/3$. A_2: $u_2 = (e_1 + e_2)/\sqrt{2}$, $\lambda_1 = 1.5$, $\lambda_2 = 0.3$, $\kappa_2 = 0.2$.

Restricting to rank-1 projectors $P_\theta = v(\theta)v(\theta)^\top$ with $v(\theta) = (\cos\theta, \sin\theta, 0)$, the per-state Rayleigh quotients are

$$\begin{aligned} v^\top A_1 v &= 3\cos^2\theta + 2\sin^2\theta, \\ v^\top A_2 v &= 0.9(\cos^2\theta + \sin^2\theta) + 1.2\sin\theta\cos\theta = 0.9 + 0.6\sin(2\theta). \end{aligned}$$

Averaging with weights $w_1 = w_2 = \frac{1}{2}$:

$$F(\theta) = \tfrac{1}{2}[3\cos^2\theta + 2\sin^2\theta + 0.9 + 0.6\sin(2\theta)] = \tfrac{1}{2}[3.4 + 0.5\cos(2\theta) + 0.6\sin(2\theta)]. \tag{23}$$

Setting $dF/d\theta = 0$ gives $\tan(2\theta_F^\star) = 1.2$, so $\theta_F^\star \approx 25.1^\circ$ and $F(P_F^\star) = \frac{1}{2}[3.4 + \sqrt{0.5^2 + 0.6^2}] = \frac{1}{2}(3.4 + 0.781) \approx 2.090$.

The surrogate matrix is

$$B_\lambda = \tfrac{1}{2}\Big[3\, e_1 e_1^\top + \tfrac{3}{2} \cdot \tfrac{(e_1+e_2)(e_1+e_2)^\top}{2}\Big] = \begin{pmatrix} 1.875 & 0.375 & 0 \\ 0.375 & 0.375 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{24}$$

The top eigenvector of the leading 2×2 block lies at $\theta_B^\star \approx 13.3^\circ$ from e_1. At $P_B^\star$:

$$F(P_B^\star) = \tfrac{1}{2}[3.4 + 0.5\cos 26.6^\circ + 0.6\sin 26.6^\circ] = \tfrac{1}{2}(3.4 + 0.447 + 0.268) \approx 2.058. \tag{25}$$

Regret comparison. The actual regret is

$$F(P_F^\star) - F(P_B^\star) \approx 2.090 - 2.058 = 0.032. \tag{26}$$

The coarse bound from (17) is

$$\sum_i w_i \lambda_2(A_i) = \tfrac{1}{2}(2 + 0.3) = 1.15, \tag{27}$$

which exceeds the true regret by a factor of $1.15/0.032 \approx 36$. The instance-dependent bound from Theorem 24 is computed using $\|P_F^\star u_1\|^2 = \cos^2(25.1^\circ) \approx 0.820$ and $\|P_F^\star u_2\|^2 = (\cos 25.1^\circ + \sin 25.1^\circ)^2/2 \approx 0.886$:

$$\sum_i w_i \lambda_2(A_i)(1 - \|P_F^\star u_i\|^2) \approx \tfrac{1}{2} \cdot 2 \cdot 0.180 + \tfrac{1}{2} \cdot 0.3 \cdot 0.114 \approx 0.197. \tag{28}$$

The instance-dependent bound is $0.197/0.032 \approx 6$ times the actual regret—substantially tighter than the coarse 36×, though still not tight at $P_F^\star$. The residual gap is due to the upper sandwich being itself loose pointwise. The picture: $P_B^\star \neq P_F^\star$ (separated by $\approx 11.8^\circ$), yet the actual value gap is small because $F(\theta)$ has a broad maximum. The coarse bound cannot distinguish a direction error from a value error; the instance-dependent bound partially can. Table 3 summarises.

Table 3: Weak-glue example: regret and bounds. $N = 2$, $d = 3$, $r = 1$, equal weights, A_i as in (22).

Quantity	Value
$F(P_F^\star)$	2.090
$F(P_B^\star)$	2.058
Actual regret $F(P_F^\star) - F(P_B^\star)$	0.032
Coarse bound $\sum_i w_i \lambda_2(A_i)$	1.150
Instance-dependent bound, eq. (20)	0.197
Coarse / actual	≈ 36
Instance-dependent / actual	≈ 6
$\theta_F^\star$, $\theta_B^\star$	25.1°, 13.3°

Table 4: Higher-dimensional weak-glue stress test ($d = 50$, $r = 5$, $N = 100$). The surrogate $P_B^\star$ captures most of the achievable operator-norm value at the cost of one eigendecomposition; both certification bounds substantially overestimate the true surrogate regret, with the instance-dependent bound consistently tighter.

Quantity	Value
$F(P_{\text{random}})$ (mean of 50 draws)	0.118
$F(P_B^\star)$ (one eigendecomposition of B_λ)	0.617
$F(P_F^\star)$ (Riemannian ascent on $\mathrm{Gr}(5, 50)$)	0.619
Actual surrogate regret $F(P_F^\star) - F(P_B^\star)$	0.0023
Coarse bound $\sum_i w_i \lambda_2(A_i)$	0.362
Instance-dependent bound, (20)	0.144
Trivial upper bound $\sum_i w_i \lambda_1(A_i)$	1.015

7.4. Higher-dimensional weak-glue stress test

To test the surrogate's behaviour beyond the 3×3 regime, we run a synthetic study at $d = 50$, $r = 5$, $N = 100$. The setup is reproducible by the accompanying script. We sample u_i from a mixture of $K = 10$ cluster centres in $\mathrm{span}\{e_1, \ldots, e_{10}\}$ (so the simple-top span of the u_i has effective dimension $> r$ and r-coverability fails), draw $\lambda_1(A_i) \sim U[0.8, 1.2]$ and $\lambda_2(A_i)/\lambda_1(A_i) \sim U[0.2, 0.5]$, and set $A_i = \lambda_1(A_i) u_i u_i^\top + \lambda_2(A_i) v_i v_i^\top$ with $v_i \perp u_i$. $P_B^\star$ is the top-r eigenspace of B_λ (one $d \times d$ eigendecomposition). $P_F^\star$ is approximated by Riemannian gradient ascent on $\mathrm{Gr}(r, d)$ warm-started from $P_B^\star$ plus several random restarts. A random rank-r projection baseline averages 50 uniform draws.

In this regime (Table 4), $P_B^\star$ recovers $F(P_B^\star) = 0.617$ versus $F(P_F^\star) = 0.619$ for the locally-optimised projection: the surrogate captures essentially the same operator-norm value, while requiring only one $d \times d$ eigendecomposition. A random rank-5 baseline averages 0.118, an order of magnitude smaller. The certification bounds are conservative as expected: the coarse bound 0.362 and the instance-dependent bound 0.144 both substantially exceed the actual regret 0.0023; the instance-dependent bound is consistently tighter. This is a single synthetic stress test, not a statistical law: the result supports computational plausibility of the surrogate at moderate d, not universal near-optimality of B_λ across regimes. A full statistical study would require sweeping cluster geometry, residual ratios κ_i, and weight schemes; we do not attempt one here. The takeaway is that the surrogate captures most of the achievable operator-norm value in this regime at the cost of one $d \times d$ eigendecomposition, where direct Riemannian optimisation of F on $\mathrm{Gr}(r, d)$ would be expensive.

8. Scope and Discussion

What is earned. For finite PSD families on $\mathbb{R}^d$ and a fixed projection rank r:

- Under the trace score $\mathrm{tr}(PM_t)$, a common optimal rank-r projection exists iff the family is r-coherent (Proposition 4).
- Under the operator-norm score $\lambda_{\max}(PK_tP)$, a common optimal rank-r projection exists iff the family is r-coverable (Proposition 17).
- r-coherence implies r-coverability strictly (Theorem 21).
- When $\{A_i\}$ fails r-coverability, the eigenvalue-weighted surrogate B_λ converts F into a Ky Fan eigenproblem; the surrogate value regret is bounded by $\sum_i w_i\lambda_2(A_i)(1 - \|P_F^\star u_i\|^2)$, which vanishes in three explicit regimes (rank-1 Jacobians, r-coverable case, common-shape family; Corollary 28).

The instance-dependent bound is structurally informative: it factors regret into residual spectral mass and optimizer misalignment, and attributes it pointwise to the states in the support of μ.

The regret bounds in Theorems 23–24 are conservative *certification* bounds: they certify the maximum loss from replacing a true operator-norm optimum by the Ky Fan surrogate, and identify the two mechanisms controlling this loss (residual spectral mass $\lambda_2(A_i)$ and optimiser misalignment $1 - \|P_F^\star u_i\|^2$), but they need not be sharp estimators of the true value gap, as the $d = 3$ counterexample of §7 demonstrates.

What is not earned. The results are stated for finite PSD families and finite-support regimes. Continuous regimes require replacing B_λ with the integral operator $\mathbb{E}_{x\sim\mu}[\lambda_1(J_xJ_x^\top)\, u_xu_x^\top]$ and analysing it spectrally; we have not proved a continuous-regime analogue. The operator-norm operators K_t are taken as given PSD inputs; in nonlinear response settings K_t would derive from a global rather than local Jacobian, and the rank-1-driven local maximiser characterisation (Proposition 15) need not extend. The geometry-first pooled objective $G(P) = \lambda_{\max}(P\,\mathbb{E}[J_xJ_x^\top]P)$ is mentioned implicitly via the rank-1 regime where $G = F$; a full treatment of the Jensen gap $F \geq G$ for $r > 1$ is left to a companion note. Approximate coverability is the natural next problem: given leading eigenspaces $\{E_i\} = \{E_{\max}(K_i)\}$, characterise when there exists $V \in \mathrm{Gr}(r, d)$ with $\mathrm{dist}(V, E_i) \leq \varepsilon$ for all i, and relate the covering radius to operator-norm regret. This is structurally different from approximate r-coherence (Corollary 13): trace-score regret admits a smooth, principal-angle-controlled bound because the Ky Fan trace is analytic in P, whereas operator-norm regret is governed by incidence and hitting geometry on $\mathrm{Gr}(r, d)$, with threshold behaviour when $E_{\max}(K_t)$ is multidimensional or when the family is nearly but not exactly r-hittable. A perturbation theory of r-coverability is left open.

Practical implication. Whenever a single rank-r projection must serve a family of objectives, the iff conditions of Propositions 4 and 17 are testable: r-coherence by equality of top-r eigenspaces; r-coverability by $\dim \mathrm{span}\{u_i\} \leq r$ in the simple-top case. When neither exact condition holds, the two structural quantities play different roles: $\Gamma_{\mathrm{tr}} \geq 0$ is the minimax unavoidable worst-case trace-score suboptimality (Corollary 13; $\Gamma_{\mathrm{tr}} > 0$ certifies non-coherence), while the weak-glue expression $\sum_i w_i\lambda_2(A_i)(1 - \|P_F^\star u_i\|^2)$ upper-bounds the loss from replacing $P_F^\star$ by the surrogate $P_B^\star$ (Theorem 24). The surrogate has independent computational value: it reduces $F(P)$ to a single $d\times d$ eigendecomposition of B_λ, where direct Riemannian optimisation of F on $\mathrm{Gr}(r, d)$ would be expensive. A higher-dimensional study (e.g. $d \sim 50$, $r \sim 5$, $N \sim 100$) is left to future computational work.

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